



KWAME NKRUMAH UNIVERSITY OF SCIENCE & TECHNOLOGY

KNUST-ATTENDANCE-APP (QRoll)

Comprehensive System Architecture, Page Directory & Operations Manual
Official Technical & User Documentation for Faculty, Staff and Students

Document Version:	Release Status:	Primary Audience:	Last Updated:
v2.5 (Production)	Active Academic Deployment	Lecturers, Tutors, Students & Admins	16 September 2026

1. Executive Overview & Mission

KNUST-ATTENDANCE-APP is a modern institutional platform engineered to eradicate proxy attendance, streamline classroom management, automate the 10-mark continuous assessment computation, and deliver instant real-time communication across Kwame Nkrumah University of Science and Technology. By fusing dynamic cryptographic rotating QR codes, multi-camera scanning engines, geofencing coordinates, PWA offline resilience, and cross-device Web Push notifications, the application delivers a frictionless, tamper-proof attendance experience for both educators and students.

Core Architectural Pillar	Institutional Benefit & Mechanism
Dynamic Rolling QR Codes	Tokens regenerate automatically every 10 seconds with cryptographic timestamps, preventing screenshot sharing, remote proxies, and off-campus fraudulent scans.
Dual Scanning Paradigms	Supports both Lecturer-as-Scanner (multi-camera high-speed scan with audio chime) and Student-as-Scanner (projector screen broadcast mode with live roster update).
Automated 10-Mark Assessment	Dynamically transforms attendance counts into official 10-mark university grading scale, highlighting students below the 75% examination eligibility threshold.
Cross-Device Web Push Alerts	Instant delivery of urgent announcements, lecture session alerts, and assignment deadlines directly to student smartphone lock screens even when the browser is closed.
Offline Attendance Caching	IndexedDB local persistence allows lecturers to scan hundreds of students without active Wi-Fi, automatically synchronizing upon network re-establishment.



2. Web Application Page Directory & Route Matrix

The application is structured into dedicated public portals, student self-service gateways, and lecturer control spaces:

Route Path	Target Users	Access Level	Core Functionality & Interactive Features
/	Public / All	Unauthenticated	Institutional landing screen, university branding, feature previews, direct gateway navigation to Student Portal, Tutor Login, and Manual.
/auth	Lecturers / Tutors	Public (Guest)	Secure educator authentication with Google Single Sign-On (OAuth) or email/password. Enforces lecturer role validation.
/student	Students	Index / Email	Student dashboard featuring 10s rolling personal QR token, real-time push notification opt-in & history log, attendance statistics, registered courses, and coursework submissions.
/manual	All Users	Public	Interactive documentation, searchable guide, step-by-step guides, troubleshooting FAQs, and PDF user manual download.
/check-in	Students	Self Check-In	Student-facing check-in screen for entering session PIN codes or scanning projected lecture room QR tokens with geofencing validation.
/portal/\$token	Students	Public Token	One-click enrollment link allowing students to self-enroll into specific course rosters without manual instructor data entry.
/dashboard	Lecturers / Staff	Authenticated	Real-time operational dashboard with active session status, daily attendance statistics, quick-action cards, recent check-in feeds, and course summaries.
/scan	Lecturers / TAs	Authenticated	Live high-speed multi-camera scanner supporting front/back cameras, continuous scan loop, audible chime, duplicate protection, and manual index fallback.
/sessions	Lecturers	Authenticated	Session hub to launch new lecture attendance sessions, customize geofencing radius, activate PIN mode, project dynamic rolling QR codes, and view live rosters.
/courses	Lecturers / Staff	Authenticated	Course catalog management: create/edit course codes, credit hours, departments, semesters, view class rosters, and track overall attendance percentages.
/students	Lecturers / Staff	Authenticated	Master student directory with multi-field search, index number filtering, profile editing, and bulk roster import via Excel (.xlsx) and CSV.
/departments	Administrators	Authenticated	University organizational tree: configure faculties, colleges, academic departments, and assign courses accordingly.
/semesters	Administrators	Authenticated	Academic calendar setup: set academic years, term dates, and switch the system-wide active semester.
/announcements	Lecturers / Staff	Authenticated	Publish emergency, department, or course-specific announcements with automatic Web Push dispatch to student devices.
/assignments	Lecturers & Tutors	Authenticated	Coursework distribution hub: set titles, descriptions, due dates, course attachments, and inspect/grade student digital submissions.
/reports	Lecturers / HODs	Authenticated	Continuous assessment gradebook: auto-calculates 10-mark scores, highlights <75% attendance disqualifications, and exports to formatted Excel, CSV, and official PDF.
/history	Lecturers	Authenticated	Immutable chronological audit log of all historical scan events with timestamps, student details, and verification methods.
/settings	Lecturers / Staff	Authenticated	Personal profile configuration, push notification subscription status, registered physical scanner devices, and security credentials.



3. Core Feature Breakdown & Engineering Mechanics

3.1 Dynamic Rolling Security QR Tokens (Anti-Proxy Engine)

Security

Static QR codes can be easily captured via screenshot and messaged to absentee colleagues. KNUST-ATTENDANCE-APP solves this via rolling cryptographic tokens that regenerate every 10 seconds with high-entropy randomized nonce salts and unix timestamps. When scanned, the system verifies the token's active window; expired or forward-shared screenshots are rejected instantly.

3.2 High-Speed Multi-Camera Scanning Engine

Performance

Built with zxing-js and HTML5 video streaming, the scanner enables lecturers and teaching assistants to scan 50+ students per minute. Features include instant audio chime confirmation, memory-resident duplicate prevention (ignoring accidental double-scans), environmental camera toggling, torch/flashlight controls for dim halls, and manual index search fallback.

3.3 Cross-Device Web Push Notifications & In-App History Feed

Real-Time

Students can subscribe their mobile phones, tablets, or laptops to native Web Push Notifications via VAPID service workers. Lecturers broadcasting urgent announcements, assignment deadlines, or class session cancellations trigger instantaneous background OS alerts on student devices even when the browser is entirely closed. Every notification is stored permanently in the student's Notification History log.

3.4 Geofencing & GPS Radius Classroom Validation

Integrity

Lecturers can lock attendance sessions to a specific lecture hall by specifying target GPS coordinates and an allowable radius (e.g., 50 meters around Great Hall or College of Science). When students submit attendance from their personal devices, the browser geolocation API confirms physical presence inside the designated radius.

3.5 10-Mark Continuous Assessment & 75% Exam Eligibility Engine

Academic

In accordance with KNUST academic statutes requiring 75% attendance for end-of-semester examination clearance, the platform automatically evaluates each student's attendance percentage. It assigns a standardized proportional score out of 10 marks and visually flags at-risk students who fall beneath the clearance threshold with warning badges.

3.6 Offline Caching & Automated Cloud Synchronization

Reliability

Ghanaian lecture halls occasionally face intermittent cellular or Wi-Fi coverage. The application utilizes IndexedDB local caching to record and timestamp student check-ins offline. As soon as connectivity is restored, the local queue automatically pushes entries to Cloud Firestore without data loss or duplicate increments.



4. End-to-End Operational Workflows

Workflow A: Lecturer Classroom Attendance Session

Step	Action & Procedure	Expected Result & Tips
1	Sign in at /auth via Google or Email/Password.	Opens the Tutor Dashboard displaying summary metrics, active courses, and recent attendance sessions.
2	Navigate to 'Sessions' and click 'Create Session'. Select course, title, and optional Geofence.	System creates an active session document with a unique session ID and dynamic security tokens.
3	Choose Mode: Click 'Launch Projector QR' OR open the 'Live Scanner' (/scan).	Projector Mode displays a huge rolling QR on the hall screen; Scanner Mode turns the lecturer's phone into a high-speed scanner.
4	Scan student personal QR codes OR let students scan the projector screen.	Audio chime confirms every record; duplicates are blocked; live count updates in real time on the dashboard.
5	Click 'End Session' once the lecture concludes.	Closes attendance session, locks submissions, and calculates cumulative attendance stats.
6	Navigate to 'Reports' to view attendance gradebook and export to Excel, CSV, or PDF.	Generates clean university grade sheets with 10-mark continuous assessment scores and exam eligibility flags.

Workflow B: Student Onboarding, Check-In & Push Notification Setup

Step	Action & Procedure	Expected Result & Tips
1	Visit /student. If visiting for the first time, click the 'Sign Up' tab.	Presents the university verification form requesting Index Number and registered email.
2	Enter Index Number, registered email, and create a 6+ character password. Click 'Sign Up'.	Instantly verifies enrolment in university database, secures credentials, and logs into student portal.
3	Click 'Enable Mobile Push Notifications' banner on the student portal header.	Prompts browser notification permission; registers device VAPID token to receive instant broadcast alerts.
4	Present the live rotating QR badge to the lecturer's camera scanner during class.	Token refreshes every 10s. Lecturer's device chimes and marks student present instantly.
5	Alternative: If lecturer is projecting, open /check-in, scan screen QR or enter 6-digit PIN.	Verifies geofence location and logs attendance directly from the student's mobile device.
6	Inspect Course Roster & Submissions in the student dashboard.	Shows attendance percentage, Continuous Assessment score (out of 10), exam status, and assignment tasks.



5. Frequently Asked Questions & Troubleshooting Guide

Scenario / Question	Root Cause & Resolution Step
Camera not opening in Scanner (/scan)?	Ensure browser permissions for camera are granted (Settings -> Site permissions -> Camera -> Allow). On iOS Safari, ensure no other background app has locked the camera. You can toggle between front and back cameras via the dropdown.
Student QR code not scanning?	Request the student increase their smartphone screen brightness to maximum. Clean any smudges on the screen. If issues persist, the lecturer can type the student's index number directly into the manual check-in bar in the scanner.
Can students scan from their hostel or off-campus?	No. Dynamic projector session QR codes rotate continuously. Furthermore, when geofencing is enabled on the session, any attempt to check in beyond the lecture venue radius is blocked with a location mismatch notice.
What happens if there is no internet in the lecture hall?	The webapp is an offline-capable PWA. The scanner records and queues scans in browser storage (IndexedDB). Once the lecturer connects to campus Wi-Fi or cellular data, all queued records synchronize automatically.
How do students receive push notifications on phone?	The student must sign in on their smartphone browser (Chrome, Edge, Safari, Firefox), click 'Enable Notifications' on the portal banner, and tap 'Allow'. The device will receive alerts even when the browser is minimized or the screen is locked.
Student forgot their portal password?	The student visits /student, selects the 'Reset Password' tab, enters their index number and registered email, and immediately chooses a new password without requiring administrator intervention.
Can a student share their account with multiple phones?	The application features Device Registration tracking under Settings. Unusual multi-device activity triggers verification safeguards to prevent credential sharing.
How is the Continuous Assessment (CA) grade computed?	Calculated as (Total Sessions Attended / Total Conducted Sessions) * 10, rounded to one decimal place. If a student falls below 75% total attendance, they are marked as 'Disqualified from Examination'.

Official Academic Support & Departmental Enquiries

For technical support, bulk student roster imports, department onboarding, or feature enhancements, contact the Academic Computing & Quality Assurance Directorate, Kwame Nkrumah University of Science and Technology. Access the live system at your institution's assigned domain.